

Exp 5.1

```
LAB_EXP_DSP
Editor - C:\Users\abhi\Desktop\LAB_EXP_DSP\exp5_1.m
exp5_1.m
1  clc; clear; close all;
2  %% --- Input Parameters ---
3  fp = 1000; % High-pass cutoff frequency (Hz)
4  fs = 8000; % Sampling frequency (Hz)
5  N = 50; % Filter order
6
7  wc = fp/(fs/2); % Normalized cutoff (0-1)
8
9  %% --- 1. FIR High Pass using Rectangular Window ---
10 b_rect = fir1(N, wc, 'high', rectwin(N+1));
11 disp('--- High-Pass Coefficients (Rectangular Window) ---');
12 disp(b_rect);
13
14 %% --- 2. FIR High Pass using Hamming Window ---
15 b_hamm = fir1(N, wc, 'high', hamming(N+1));
16 disp('--- High-Pass Coefficients (Hamming Window) ---');
17 disp(b_hamm);
18
19 %% --- 3. FIR High Pass using Hanning Window ---
20 b_hann = fir1(N, wc, 'high', hannwin(N+1));
21 disp('--- High-Pass Coefficients (Hanning Window) ---');
22 disp(b_hann);
23
24 %% --- Frequency Response Plots ---
25 figure;
26 freqz(b_rect,1);
27 title('High-pass FIR using Rectangular Window');
28 figure;
29 freqz(b_hamm,1);
30 title('High-pass FIR using Hamming Window');
31 figure;
32 freqz(b_hann,1);
33 title('High-pass FIR using Hanning Window');
```

Output

--- High-Pass Coefficients (Rectangular Window) ---

-0.0089

-0.0000

0.0097

0.0143

0.0106

-0.0000

-0.0117

-0.0175

-0.0131

-0.0000

0.0148

0.0225

0.0171

-0.0000

-0.0202

-0.0314

-0.0247

-0.0000

0.0318

0.0524

0.0445

-0.0000

-0.0741

-0.1572

-0.2223

0.7407

-0.2223

-0.1572

-0.0741

-0.0000

0.0445

0.0524

0.0318

-0.0000

-0.0247

-0.0314

-0.0202

-0.0000

0.0171

0.0225

0.0148

-0.0000

-0.0131

-0.0175

-0.0117

-0.0000
0.0106
0.0143
0.0097
-0.0000
-0.0089

--- High-Pass Coefficients (Hamming Window) ---

-0.0007
-0.0000
0.0009
0.0016
0.0015
-0.0000
-0.0024
-0.0044
-0.0039
-0.0000
0.0060
0.0103
0.0088
-0.0000
-0.0128
-0.0217
-0.0184
-0.0000
0.0268
0.0464
0.0410
-0.0000
-0.0725

-0.1567
-0.2240
0.7493
-0.2240
-0.1567
-0.0725
-0.0000
0.0410
0.0464
0.0268
-0.0000
-0.0184
-0.0217
-0.0128
-0.0000
0.0088
0.0103
0.0060
-0.0000
-0.0039
-0.0044
-0.0024
-0.0000
0.0015
0.0016
0.0009
-0.0000
-0.0007

--- High-Pass Coefficients (Hanning Window) ---

-0.0000

-0.0000

0.0003

0.0008

0.0009

-0.0000

-0.0020

-0.0038

-0.0035

-0.0000

0.0057

0.0100

0.0087

-0.0000

-0.0127

-0.0216

-0.0183

-0.0000

0.0267

0.0464

0.0410

-0.0000

-0.0726

-0.1568

-0.2243

0.7500

-0.2243

-0.1568

-0.0726

-0.0000

0.0410

0.0464

0.0267

-0.0000

-0.0183

-0.0216

-0.0127

-0.0000

0.0087

0.0100

0.0057

-0.0000

-0.0035

-0.0038

-0.0020

-0.0000

0.0009

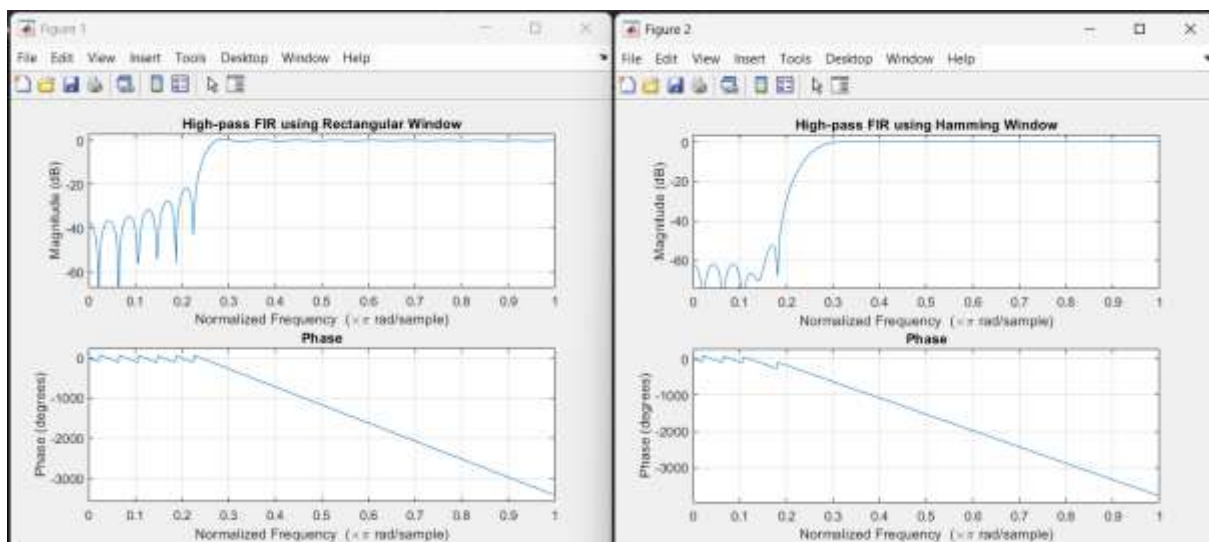
0.0008

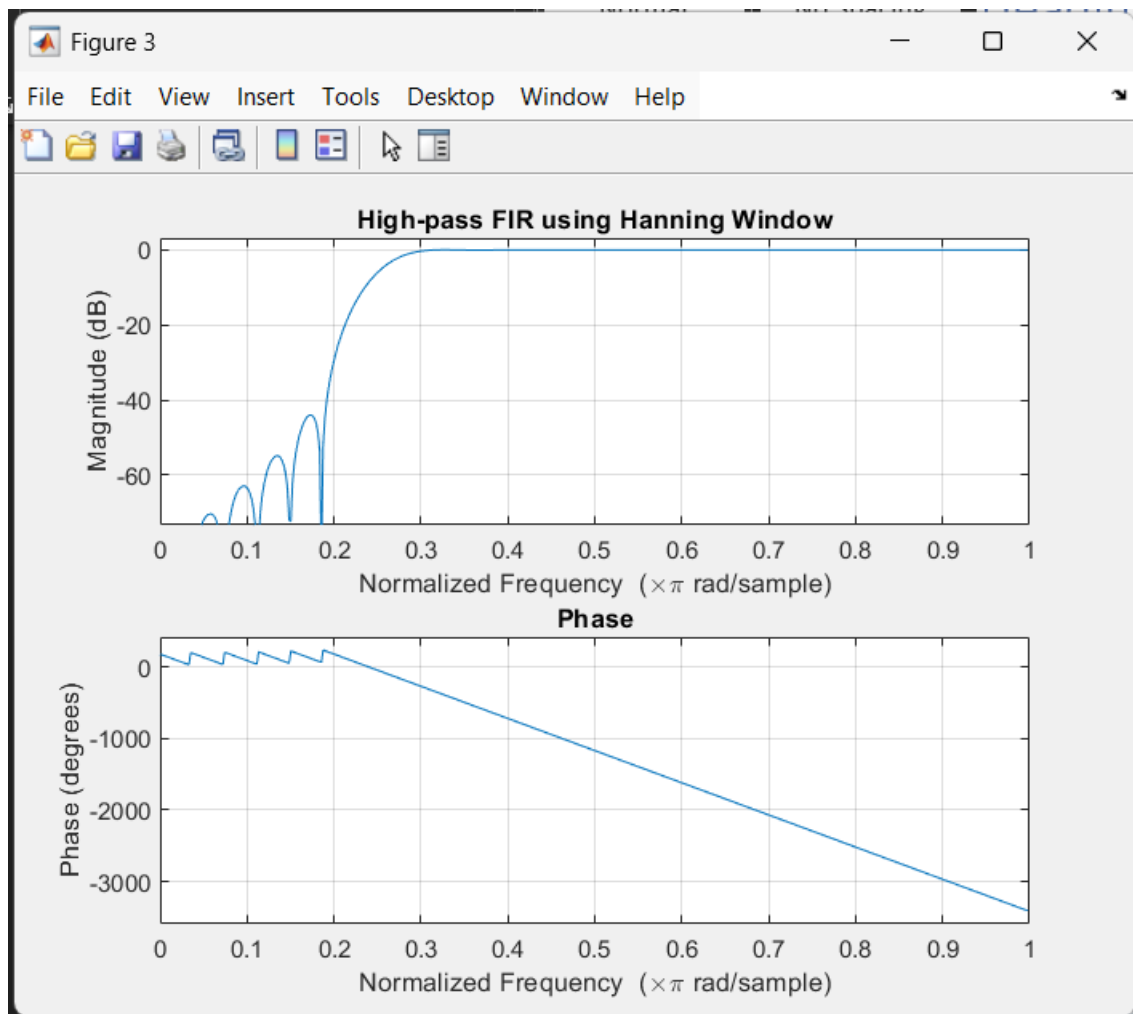
0.0003

-0.0000

-0.0000

>>





Exp 5.2

```
Editor - C:\Users\abhiz\Desktop\LAB_EXP_DSP\exp5_2.m
exp5_1.m exp5_2.m +
1 clc; clear; close all;
2 %% --- Input Parameters ---
3 fp = 1000; % High-pass cutoff frequency (Hz)
4 fs = 8000; % Sampling frequency (Hz)
5 N = 50; % Filter order
6 wc = fp/(fs/2); % Normalized cutoff (0-1)
7 %% --- FIR High Pass using Hamming Window ---
8 b_hamm = fir1(N, wc, 'high', hamming(N+1));
9
10 disp('--- High-Pass Coefficients (Hamming Window) ---');
11 disp(b_hamm);
12 %% --- Frequency Response Plots ---
13 figure;
14 freqz(b_hamm,1);
15 title('High-pass FIR using Hamming Window');
16 title('High-pass FIR using Hanning Window');
```

Output

--- High-Pass Coefficients (Hamming Window) ---

-0.0007
-0.0000
0.0009
0.0016
0.0015
-0.0000
-0.0024
-0.0044
-0.0039
-0.0000
0.0060
0.0103
0.0088
-0.0000
-0.0128
-0.0217
-0.0184

-0.0000

0.0268

0.0464

0.0410

-0.0000

-0.0725

-0.1567

-0.2240

0.7493

-0.2240

-0.1567

-0.0725

-0.0000

0.0410

0.0464

0.0268

-0.0000

-0.0184

-0.0217

-0.0128

-0.0000

0.0088

0.0103

0.0060

-0.0000

-0.0039

-0.0044

-0.0024

-0.0000

0.0015

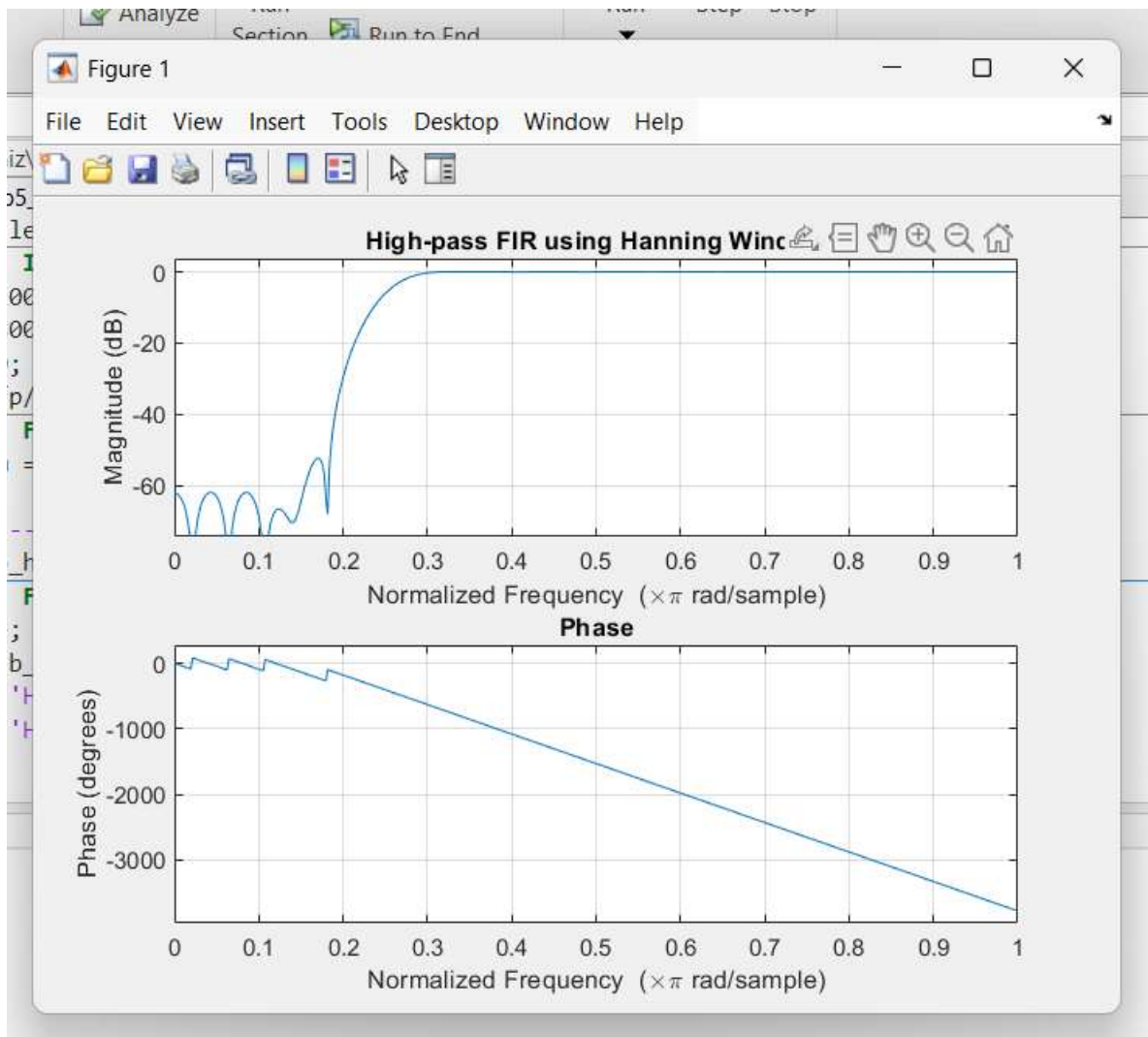
0.0016

0.0009

-0.0000

-0.0007

>>



Exp 5.3

```

Editor - C:\Users\abhiz\Desktop\LAB_EXP_DSP\exp5_3.m
exp5_1.m  exp5_2.m  exp5_3.m  +
1      clc; clear; close all;
2
3      %% --- Input Parameters ---
4      fp = 1000; % High-pass cutoff frequency (Hz)
5      fs = 8000; % Sampling frequency (Hz)
6      N = 50; % Filter order
7
8      wc = fp/(fs/2); % Normalized cutoff (0-1)
9      %% --- FIR High Pass using Hanning Window ---
10
11     b_hann = fir1(N, wc, 'high', hanning(N+1));
12     disp('--- High-Pass Coefficients (Hanning Window) ---');
13     disp(b_hann');
14     %% --- Frequency Response Plots ---
15     figure;
16     freqz(b_hann,1);
17     title('High-pass FIR using Hanning Window');

```

Output

--- High-Pass Coefficients (Hanning Window) ---

```

-0.0000
-0.0000
0.0003
0.0008
0.0009
-0.0000
-0.0020
-0.0038
-0.0035
-0.0000
0.0057
0.0100
0.0087
-0.0000
-0.0127
-0.0216

```

-0.0183

-0.0000

0.0267

0.0464

0.0410

-0.0000

-0.0726

-0.1568

-0.2243

0.7500

-0.2243

-0.1568

-0.0726

-0.0000

0.0410

0.0464

0.0267

-0.0000

-0.0183

-0.0216

-0.0127

-0.0000

0.0087

0.0100

0.0057

-0.0000

-0.0035

-0.0038

-0.0020

-0.0000

0.0009

0.0008

0.0003

-0.0000

-0.0000

>>

